

Baden Albert Dominic Paiva

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EDUCATION

American University of Sharjah

Bachelor of Science in Computer Science

Sharjah, UAE

Sep 2023 – Present

- Chancellor's List Awardee (2023 - 2025)
- Member of the Tau Beta Pi Engineering Honors Society

The Indian High School Dubai

High School

Dubai, UAE

Apr 2009 – March 2023

- Yallaschools Middle East Education High Achievers' Award (2023)

EXPERIENCE

Undergraduate Lab Assistant

American University of Sharjah

Sep 2025 – Present

Sharjah, UAE

- Assisted students in debugging and developing x86 assembly programs, providing guidance on low-level programming, registers, memory addressing, and control flow.
- Reinforced computer architecture fundamentals through one-on-one instruction during lab sessions
- Stepped in to lead lab sections independently during instructor absences, ensuring continuity of instruction

Volunteer Trainer for Coding & Robotics

The Indian High School

May 2022 – July 2022

Dubai, UAE

- Facilitated hands-on activities in Tinkercad to build coding intuition and logical problem solving skills
- Taught app development using MIT App Inventor using a project-based learning approach to help them translate ideas into functional applications
- Assessed student progress through project demonstrations, supporting skill development and confidence

PROJECTS

Machine Learning–Based Heart Disease Prediction | *Python, pandas, NumPy, Seaborn, Matplotlib, scikit-learn*

- Developed an end-to-end machine learning pipeline using Python to preprocess 918 patient records, implementing missing value imputation, categorical encoding, and normalization
- Applied feature selection techniques using scikit-learn and performed PCA-based dimensionality reduction to create 3 dataset variants for comparative analysis
- Implemented and trained various classifiers with hyperparameter tuning across multiple parameters using 5-fold stratified cross-validation to optimize model generalization
- Visualized model performance metrics using Matplotlib and Seaborn, creating ROC curves, confusion matrices, and hyperparameter tuning plots to evaluate metrics across all models

Unitrade - E-Commerce Web Platform | *Java, MySQL, HTML/CSS, JavaScript, Bootstrap, Jakarta Servlets*

- Developed full-stack peer-to-peer marketplace using Java Servlets, MySQL, HTML/CSS, JavaScript, and Bootstrap, enabling students to buy and sell second-hand items with user authentication and listing management
- Implemented 15+ RESTful servlets handling CRUD operations for user accounts and product listings, with DAO classes for database interactions and JSON-based data exchange between frontend and backend
- Built a dynamic frontend with responsive navigation, product search and filtering by category/keywords and user dashboard for managing personal listings and sold status
- Integrated Jakarta Mail API to create automated email notification system that connects buyers and sellers

Sudoku Solver Web Application | *Python, Flask, WebSockets, Algorithm Design*

- Developed and deployed interactive Sudoku solver web application using Flask and WebSockets, hosted on Render with real-time board solving
- Implemented backtracking algorithm with recursive depth-first search to solve Sudoku puzzles, validating the game's constraints for efficient puzzle resolution
- Built Sudoku board generation system creating puzzles by strategically removing cells from completed valid boards

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (MySQL), JavaScript, HTML/CSS, x86 Assembly

Developer Tools: GitHub, VS Code, Visual Studio, Eclipse, Figma, Netlify, Render

Libraries: pandas, NumPy, Matplotlib, Seaborn, scikit-learn, jQuery, Bootstrap, Flask, slick.js